

Name: [REDACTED]

Great work!

Date: _____

Structure and Properties Unit TestPart A: Multiple Choice (15 marks - on D2L)Part B: Thinking (20 marks)

1. For each of the following molecules, do the following: (8 marks)

- Draw the Lewis structure and indicate the geometric shape as predicted by the VSEPR Theory
- Determine if the molecule is polar or non-polar
- Indicate the hybridization around the central atom

Compound	Lewis Structure (Include Bond Dipoles)	VSEPR name (molecular geometry)	Polar or Non-Polar	Hybridization on Central Atom
7.5 BF_3		trigonal planar ✓	nonpolar ✓	sp^2 ✓
SF_4		seesaw / sawhorse ✓ net dipole	polar ✓	sp^3d ✓

② For the electron indicated below, give the value of the 4 quantum numbers. Be sure to name the quantum numbers in your answer. (4 marks)

4

Energy ↑

2p $\boxed{\uparrow\downarrow} \boxed{1} \boxed{1} \boxed{1}$

2s $\boxed{\uparrow\downarrow}$

1s $\boxed{\uparrow\downarrow}$

oxygen: 2, 1, -1, $-\frac{1}{2}$

n = principle quantum # (size + energy) 2 ✓

l = secondary quantum # (orbital shape) 1 ✓

m_l = quantum orientation # (orientation) -1 ✓

m_s = quantum spin # (electron spin) $-\frac{1}{2}$ ✓

③ Determine the **identity** of the following elements. Indicate how many electrons were added or removed for each. The electron configuration given is for the ion. (3 marks)

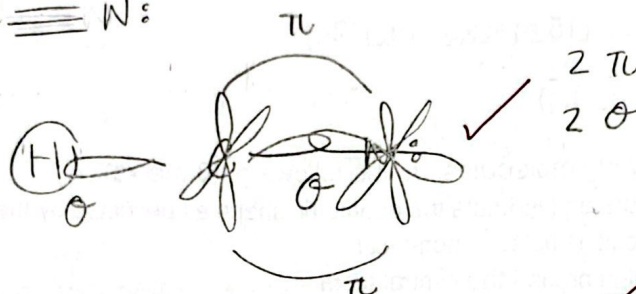
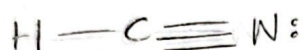
a) The element that forms an ion with a -3 charge and has the electron configuration $[\text{He}]2s^22p^6$.

there are 3 electrons added for a -3 charge $\hookrightarrow [\text{He}]2s^22p^3$
nitrogen (N) is the original element

b) The element that forms an ion with a +4 charge and has the electron configuration $[\text{Ar}]3d^5$.

there are 4 electrons removed for a +4 charge $\hookrightarrow [\text{Ar}]4s^23d^5$
Iridium (Ir) is the original element Cobalt

4. For a molecule of HCN, complete the following: (5 marks total)
a) Draw the molecular orbital bonding diagram. (2 mark)



only need to draw lone pairs on central atoms for bonding diagram

b) What is the hybridization of the central carbon atom? sp (1 mark)

c) Predict the VSEPR shape. linear (1 mark)

d) Identify any sigma (σ) or pi (π) bonds by labelling them on your orbital bonding drawing in part (a). (1 mark)

2 π 2 σ

Part C: Communication (10 marks)

1. Use the concepts of VSEPR shape and intermolecular forces, determine which has the higher boiling point - OF_2 or BH_3 . Complete the chart below. (4 marks)

Compound	Molecular VSEPR shape	Intermolecular Forces
OF_2	bent	hydrogen bonding dipole-dipole London dispersion
BH_3	trigonal planar	London dispersion
Which has the higher boiling point <u>and</u> <u>explain</u>	OF_2 has a higher boiling point because its intermolecular bonds include <u>dipole-dipole</u> <u>H-bonds</u> , which are far stronger than London dispersion, so it needs more energy to break apart the H-bonds, leading to a higher boiling point	

need Hydrogen for H-bonding

2. What is wrong with each of the following notations? Briefly explain the error. (2 marks)

a) 3, 2, 0, -1

the quantum spin number (m_s) is equal to -1 when it can only have the values of $\pm \frac{1}{2}$

b) $[\text{Ar}] 4s^2 4p^3$

After the $4s^2$ orbital, there is no $3d$ orbital when there should be, before the $4p$ orbital. this violates Aufbau's rule as electrons must fill orbitals from low to highest energy order

3. Choose either a or b. State and explain the similarities and differences in the properties of the solids. Be sure to indicate the **type of bonding** and **name the solid structure**. (4 marks)

a) Solid Na and Na₂O

OR

b) Graphite and diamond.

3.5

Na	similarities	Na ₂ O
- metallic bonding (delocalized e ⁻)	- have metal elements	- ionic bonding (ΔEN: 3.5 - 0.9 = 2.6)
- metallic crystal structure (soft metal)	- have high strength (not always)	- ionic crystal structure
- good conductor of electricity (solid)	- have high malleability (Na ₂ O is brittle)	- high boiling point
- malleable	- tightly bonded	- good conductor of electricity in water
	- new strong intermolecular forces	

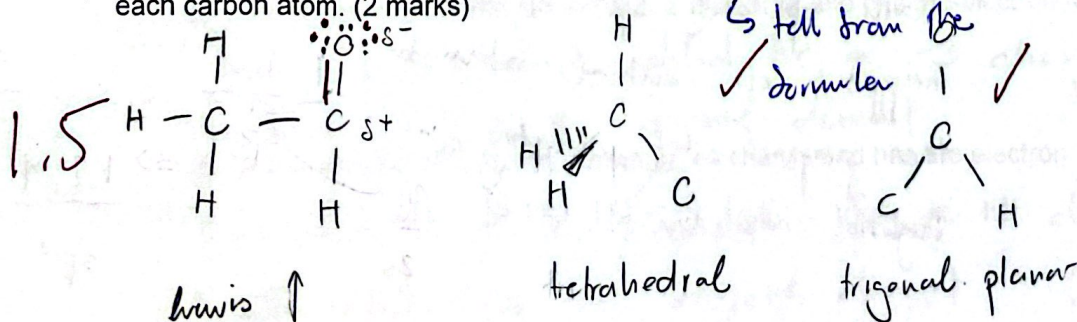
Part D: Application (10 marks)

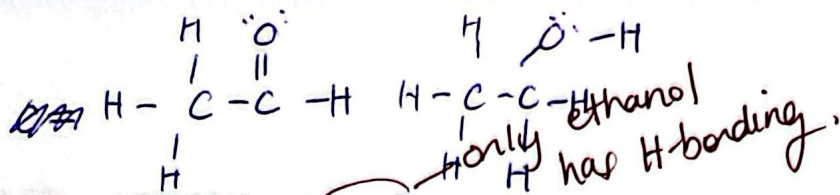
1. Compare and contrast the solids carbon dioxide (CO₂) and sodium chloride (NaCl) by completing the following chart: (4 marks)

3

Property	carbon dioxide	sodium chloride
Structure (type of solid)	molecular compound crystal	ionic crystal
Melting/ Boiling Point	relatively low melting/ boiling point	relatively high melting/ boiling point
Electrical Conductivity	not a good conductor	not a good conductor solid good conductor in water
Malleability/Ductility	low malleability/ductility because the bonds are very strong little hardness	highly malleability/ductility because the ionic bonds keep the compound bonded tightly Hard and Brittle

2. a) Draw the Lewis structure for ethanal (CH₃COH). Predict the shapes individually around each carbon atom. (2 marks)



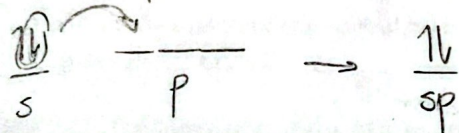


b) Ethanal (CH_3COH) and ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) have very similar structures and molecular masses. Explain why the boiling point of ethanal is 20°C and ethanol is 78°C . (2 mark)

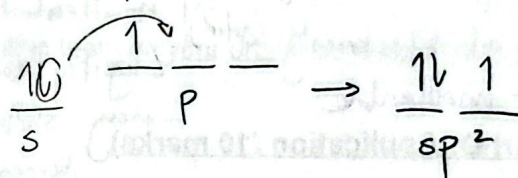
they both have all intermolecular forces, including CDF, dipole-dipole, and hydrogen bonding. However, ethanol ($\text{CH}_3\text{CH}_2\text{OH}$) is a larger molecule, which means it has more bonds and atoms that require more energy to break apart, resulting in a higher boiling point.

3. Compare and contrast sp hybridization and sp^2 hybridization. For each type, give an example of a molecule in which it is found. (2 marks)

sp hybridization



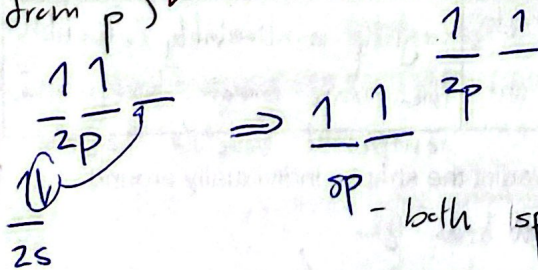
sp^2



- found in compounds with linear shape

- ex. HCN

- both e^- originally found in s orbital, one transfers to p orbital $\rightarrow$ creates sp orbital (1 e^- from s , 1 e^- from p)



- found in compounds with trigonal planar shape

- ex. BH_3

- 2 electrons in s orbital, 1 electron in p orbital originally $\rightarrow$ one electron from s transfers to p to create sp^2 orbital (1 e^- from s , 2 e^- from p)

- both sp and sp^2 hybridization happens on central atom for the purpose of creating a hybrid orbital to bond

